

No Matrik									

Name: _____

Group Tutorial: _____

KOLEJ MATRIKULASI PULAU PINANG
KEMENTERIAN PENDIDIKAN MALAYSIA
PENANG MATRICULATION COLLEGE
MINISTRY OF EDUCATION MALAYSIA

PRE-PSPM 1
PROGRAM MATRIKULASI SISTEM EMPAT SEMESTER
PRE -PSPM 1
MATRICULATION PROGRAMME FOUR SEMESTER SYSTEM

SAINS KOMPUTER 1
2 JAM

JANGAN BUKA SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

ARAHAN KEPADA CALON:

Kertas ini mengandungi 6 soalan.
Jawab **semua** soalan di ruang yang disediakan.

INSTRUCTION TO CANDIDATE:

This booklet consists of 6 questions. Answer **all** questions in the space provided.

Untuk Kegunaan Pemeriksa		
No. Soalan	Pemeriksa	KKP/KP
1		
2		
3		
4		
5		
6		
JUMLAH		

Kertas soalan ini mengandungi 10 halaman bercetak.
This booklet consists 10 printed pages.

1. (a) *“Refers to a dynamic and visually engaging component that involves the movement or transformation of images, graphics, or other visual elements.”*

(i) Describe the multimedia elements in statement 1(a).

[2 marks]

(ii) Explain the purpose of multimedia elements in statement 1(a).

[2 marks]

- (b) Differentiate the dimension aspect of multimedia representation between **M** and **N** in Figure 1.

[2 marks]

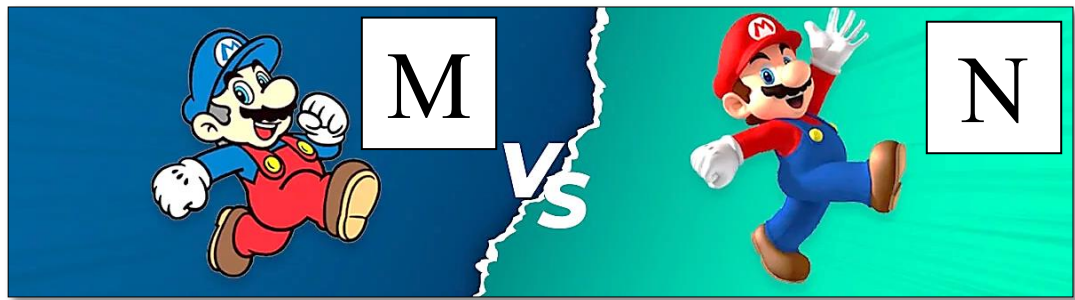


Figure 1

M: _____

N: _____

2. (a) A multimedia designer can contribute to various organizations or industries that use engaging content for entertainment, information, or education. The creation and development process of multimedia products is crucial for ensuring their quality.

(i) There are 3 phases involved in designing and developing multimedia. Briefly describe the third phase of the multimedia design.

[2 marks]

- (ii) Identify the activity in multimedia phases based on the description below.
[4 marks]

Description	Activity
Isya is incharge of converting non digital materials into digital materials	
Aizal needs to ensure that the multimedia product is suitable in any type of devices such as laptops, tablets and personal computers.	
Una needs to identify how the user will interact with the interface of the multimedia product.	
Haya has to make sure that the end product is at par with the storyboard.	

- (iii) An authoring tool is designed to facilitate users to produce interactive and multimedia-rich materials without extensive programming knowledge.

Identify and differentiate the authoring tools based on description in Table 1.

[4 marks]

A <i>The tool allows authors to synchronize and control the timing of different elements within the content, such as text, images, audio, and video.</i>	B <i>The tool may provide templates or a blank canvas for users to arrange and design content and is often used for designing websites, digital documents, or presentations.</i>
A: _____	B: _____

Table 1

- (iv) Explain hypermedia that is shown in Figure 2.

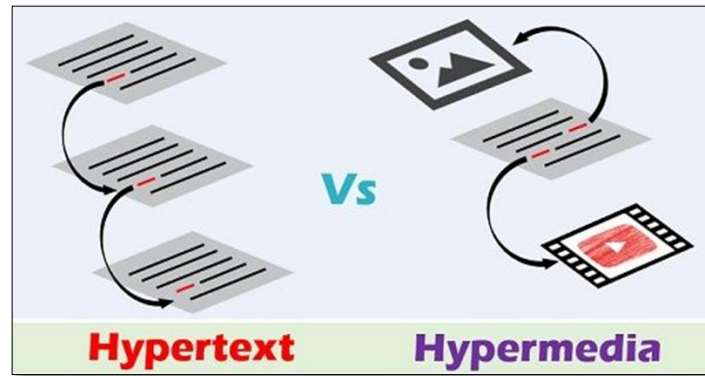


Figure 2

[2 marks]

- (b) Sending a resume to a future employer involves a thoughtful and professional approach.

- (i) Differentiate the method for delivering a resume to a potential employer.

[4 marks]

	Method 1	Method 2
Platform		
Speed		

3. An early attempt at automating manual procedures before database systems existed was the use of file-based systems.

- (a) Differentiate between file processing system and database system based on data redundancy factor.

[2 marks]

File Processing System	Database System

(b) State two (2) drawbacks of database system.

[2 marks]

(c) “Digital data is unaltered and that only those with permission can access or alter it”

Explain the concept of data integrity.

[2 marks]

4. Any defined item in a database used for data storage or reference is called a database object.

(a) Explain the function of database objects below.

[2 marks]

Database Object	Functions
Form	
Query	

(b) “A way of summarizing and presenting data. This could be for a variety of purposes such as showing trends, summaries, or detailed data.”

State the database object described above.

[1 mark]

(c) State the function of database object in 4(b).

[1 mark]

5. (a) A subtle guide to location through signals from orbiting satellites.

(i) Identify and describe the network's use based on the description given above.

[2 mark]

(b) Networks play a fundamental role in connecting people, devices, and systems, enabling communication, resource sharing, and diverse applications.

(i) State one (1) network use in entertainment.

[1 mark]

(ii) Describe your answer in 5(b)(i).

[2 marks]

(c) Different types of media may be used within the same network to meet specific requirements for different segments or applications.

(i) State one (1) difference between wired and wireless networking media.

[4 marks]

	Wired	Wireless
Speed of data transmission		
Installation cost		

(ii)

Give one (1) example of each of the networking media.

[2 marks]

Wired:

Wireless:

(d) Diagram 1 below shows a basic network diagram.

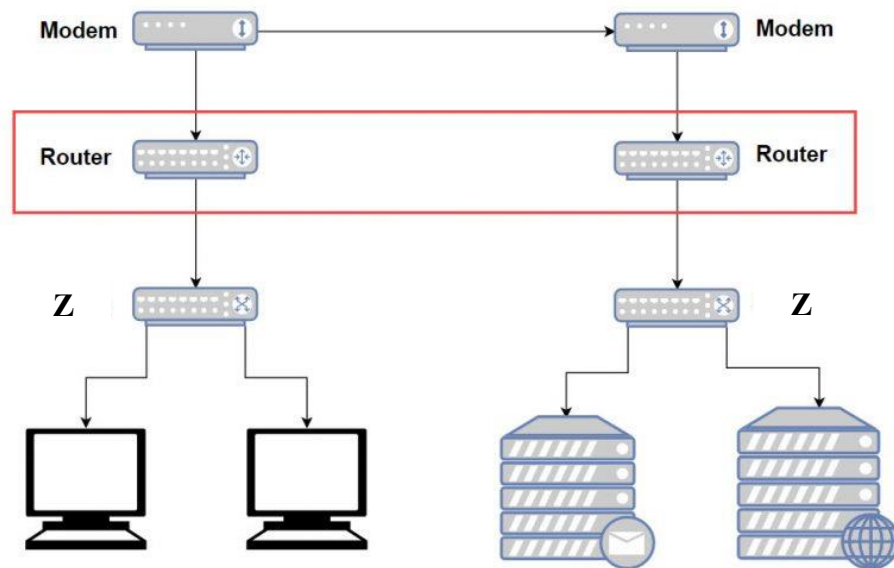


Diagram 1

(i) Describe the function of router in the network above.

[2 marks]

(ii) Give one (1) reason why modem is used in the network above.

[2 marks]

(iii) Name **Z** network device that is used to build a network besides modem and router.

[1 mark]

(iv) Describe the function of the device in 5(d)(iii).

[2 marks]

6. (a) Your supervisor at an electronics store asked you to find out what the top 5 smartwatches are this year. They want to ensure that the store has these popular models in stock for the upcoming marathon events. In order for you to find out that information, you turn on a computer and get ready to conduct a search on the internet.

(i) Identify the most suitable type of search tool to be used in the scenario. [1 mark]

(ii) Describe the type of search tool mentioned above. [2 marks]

- (b) Search strategies will help searching process be more efficient and effective. Identify the search strategy used based on description given.

(i) Search strategy allows you to search for results that include both search terms. [1 mark]

(ii) Searching involves using quotation marks to find exact matches. [1 mark]

(iii) Search strategy allows you to specify which attribute of a document you want to search within, such as the title, author, or publication date. [1 mark]

- (c) Which search strategies most efficient?
Explain your answer.

[2 marks]

- (d) The majority of Internet users engage in a variety of ordinary activities online, ranging from everyday duties and social planning to personal leisure interests.

- (i) Identify the types of web activities based on the descriptions given below. [3 marks]

Descriptions	Web Activity
Encompass streaming movies and music to engage in multiplayer video games and immersive digital leisure landscape.	
Facilitate remote learning, virtual classrooms, interactive assignments, and collaborative discussions through digital platforms.	
Involve users creating, sharing, and engaging with content, fostering virtual connections and communities' relationship-building.	

Table 2

- (ii) Describe web activity of e-Commerce. [2 marks]

- (e) Internet users have to apply precaution when using digital technology for socializing, relaxation, and educational purposes, distinguishing between acceptable and undesirable content.

- (i) Identify the issues that these internet users may be experiencing. [3 marks]

Scenario	Issue
Our online activities are discreetly tracked by cookies, allowing advertisers to target us with personalized ads.	
Government restricts access to certain websites and platforms, stifling free expression and limiting the flow of information among citizens.	
An employee accidentally sent a confidential client email to the wrong person.	

- (ii) State the methods to improve privacy when using the Internet.

[2 marks]

	Method to improve privacy
Spyware	
Unauthorize access	

- (f) A common application of the Internet of Things (IoT) in home automation is the creation of smart homes.

- (i) Explain the positive impact of IoT in home automation.

[2 marks]

END OF QUESTIONS